

Case Study Rubric

DS 4002 – Spring 2026 - Tiannah Griffith

Due: TBD

**Submission format: Upload link to GitHub repository on Canvas that includes all content.
Individual Assignment**

General Description: Submit on Canvas a link to your GitHub repository that contains all required content.

Preparatory Assignments – Basic knowledge of Python, GitHub, and introductory machine learning concepts.

Why am I doing this? This case study gives you an opportunity to work with real U.S power outage data and practice a complete data science workflow, from cleaning and exploration to feature engineering, modeling, and interpretation. Beyond reproducing an existing analysis, you will extend it by engineering new features and evaluating whether they improve model performance. By reproducing and extending the work of a previous team, you will develop a skill that is critical in professional settings: picking up where someone else left off, understanding their decisions, and building on them meaningfully.

- Course Learning Objective: Develop and evaluate predictive models using real-world data; practice critical reflection on modeling decisions

What am I going to do? You will work with the Event-Correlated Outage Dataset from Data.gov, which contains over 77,000 records of power outage events across the United States in 2023. You will clean the data, conduct new exploratory data analysis focused on temporal patterns, engineer at least one new time-based feature, and build three machine learning models to predict the hour of day a power outage begins. You will compare your models using R^2 , Mean Absolute Error, and Root Mean Squared Error and determine if your new features helped. Deliverables include:

- A clean, reproducible GitHub repository – containing all code, data, outputs, written analysis, and references

Tips for success:

- Start by thoroughly understanding the original analysis before extending it – read through the provided code and make sure you can reproduce before adding anything new.
- Let your EDA guide your feature engineering – understand your data first.
- Be patient with model run times – Random Forest and XGBoost can take a while on a dataset this large
- Keep your code commented and organized as you go – do not wait until the end to make comments
- Don't overthink your feature – a well-justified simple feature is more valuable than a complex one you cannot explain
- Do not just report your metrics, interpret them.

How will I know I have Succeeded? You will meet expectations on this case study when you follow the criteria in the rubric below.

Category	Details
Formatting	• GitHub Repository – Submit a link to your GitHub repository on Canvas • Contents ▪ README.md ▪ Written Analysis PDF ▪ Code ▪ SCRIPTS folder ▪ DATA folder ▪ OUTPUT folder ▪ References • All data and code used in the analysis must be included in the repository • For code and data products use the appropriate format (.ipynb for notebooks, .csv for data) and PDF for written components
README.md	• An overview of what you reproduced for the case study • Briefly explain the contents of the repository
Written Analysis PDF	<u>Goal:</u> Communicate your decisions, findings, and conclusions clearly and in context. • In a few sentences, summarize the problem of predicting power outage timing and explain why it matters for utility companies and emergency responders • In a few sentences, justify the time-based feature(s) you engineered – explain what temporal pattern you found in your EDA and why you chose to capture it • Discuss your model results, identifying which model performed the best and what your evaluation metrics suggest ◦ Compare your best MAE to the original benchmark of 1.51 hours and reflect on whether your feature engineering made a difference • Acknowledge at least two limitations or sources of bias present in your dataset or models • In a few sentences, connect your findings to real-world operational decisions
Code	<u>Goal:</u> Produce clean and reproducible code. In the code you must include: • A cleaning script that preprocesses the raw dataset and saves a cleaned csv • EDA visualizations exploring temporal patterns in the data ◦ At least one new visualization exploring a temporal pattern of your choosing • A modeling notebook that: ◦ One-hot encodes categorical variables ◦ Splits and scales the data ◦ Runs Linear Regression (baseline), Random Forest, and XGBoost before and after hyperparameter tuning ◦ Includes your new engineered feature(s) in the feature matrix • A clearly formatted performance table comparing R^2, MAE, and RMSE for all three models before and after tuning • Comments throughout so a reader can follow your process
References	• List of references in IEEE citation style • You must cite the dataset and one additional source that informed your modeling approach or feature engineering decisions

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